

Justin Hsu

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Education

University of California, Davis

Bachelor of Science in Electrical Engineering

Davis, CA

2022-2026

Georgia Institute of Technology

Master of Science in Electrical Engineering

Atlanta, GA

2026-2028

Experience

RTL Design Engineer Intern

Lotus Communication Systems

Acton, MA

June 2025-September 2025

- Designed and implemented a digital Finite Impulse Response (FIR) filter in Verilog, utilizing MATLAB for the calculation of complex filter weights and coefficients.
- Implemented a complete block up/down converter (BUDC) in Verilog for deployment on Xilinx FPGAs using Vivado.
- Focused heavily on physical layout efficiency and timing closure, rewriting critical blocks to minimize multiply-and-accumulate (MAC) calculations, thereby optimizing hardware area and improving operational speed.

PCB Layout Intern

Lotus Communication Systems

Acton, MA

June 2024-September 2024

- Designed and implemented a complex mixed-signal PCB layout for a gigahertz-frequency Block Up/Down Converter (BUDC).
- Successfully isolated sensitive RF signal paths from the digital microcontroller block to minimize noise and interference.
- Optimized component placement and routing strategies (e.g., differential pairs, controlled impedance traces) to ensure signal integrity and achieve efficient use of board area in Siemens PADS and Cadence Allegro.

Projects

Handwritten Number CNN Accelerator

April 2026-June 2026

- Architected a 2-layer CNN accelerator on an Altera DE1-SoC FPGA, performing a microarchitectural trade-off analysis across weight-stationary, input-stationary, output-stationary, and row-stationary systolic array dataflows to optimize MAC computational density, localized register file reuse, and global memory bandwidth.
- Drove the complete RTL-to-GDSII physical design flow using Synopsys Design Compiler for logical synthesis and Cadence Innovus for place-and-route, leveraging the 45nm Nangate Open Cell Library to meet target timing constraints.
- Reduced power consumption via clock gating and multi-voltage domain floorplanning.

Live Matched Filter Object Detection Accelerator

January 2026 – June 2026

- Developed a pipelined, real-time Canny Edge Detection hardware accelerator in Verilog, processing a live 60 FPS streaming video input from a D8M digital camera module and driving an unbuffered VGA controller output on an Altera DE1-SoC platform.
- Evaluated architectural trade-offs for the matched filtering compute core by designing, implementing, and comparing three distinct hardware microarchitectures to perform the same spatial correlation function: a parallelized score tree, a multi-channel systolic array, and a hardware 2D-FFT block to perform computation in the frequency domain.
- Designed dedicated logic to extract bounding box coordinates from the matched filter output, streaming real-time spatial coordinates to an onboard processor for dynamic object tracking.

Heterogeneous Multicore RISC-V Processor

January 2025-March 2026

- Designed a heterogeneous RISC-V multi-core architecture in SystemVerilog, integrating a deeply pipelined, high-performance "Big" core alongside an energy-efficient "LITTLE" core mapped to a hardware-enforced cache coherence protocol.
- Authored robust, self-checking constrained-random testbenches in Cadence Xcelium to perform cycle-accurate functional verification, protocol compliance checking, and critical path timing analysis.
- Synthesized and floorplanned the multi-core design using Design Compiler and Innovus, optimizing pipeline register placement, clock tree synthesis (CTS), and routing congestion to maximize instruction throughput and frequency.

Subthreshold Minimum Energy 32-Bit Adder

January 2026-March 2026

- Engineered a low-power 32-bit Square Root Carry Select Adder (SRCSA) optimized for subthreshold operation, achieving a clock frequency of 100MHz at an ultra-low supply voltage $V_{dd} = 0.32V$
- Designed a custom cell library utilizing even and odd mirror adder topologies while strategically mixing multi-threshold devices (High Vt, Normal Vt, Low Vt) to establish a balanced subthreshold performance envelope, maintaining active sub-nanosecond delay metrics while mitigating subthreshold leakage.
- Executed full-custom transistor-level layout in Cadence Virtuoso adhering to 45nm CMOS DRC/LVS rules; performed post-layout parasitic extraction (PEX) and characterized Power-Delay Product (PDP), Energy-Delay Product (EDP), and static leakage power using Cadence Spectre and Xcelium.

Skills

Languages: SystemVerilog, Verilog, C, Python, TCL, MATLAB

ASIC/EDA Tools: Cadence Spectre, Xcelium, Innovus, Virtuoso, Allegro, Synopsys Design Compiler, Siemens PADS, Vivado, Quartus, Verilator, MAGIC, Yosys, KLayout

Technical Competencies: STA, Layout, PPA Analysis, Logic Design, Lower Power Design, HW-SW Co-design, Microsoldering